

# Smart Study Planner - Project Report

## SMART STUDY PLANNER

### A Priority-Based Study Time Allocation System

## Project Report

### Abstract

Students often find it difficult to distribute their limited study hours among various subjects with varying levels of difficulty, urgency, and grade weightage. The Smart Study Planner is a console-based Java application that helps in allocating study hours according to the difficulty level while also considering the grade weightage and the number of days left before each examination. This project report discusses the motivation, design, working algorithm, implementation, and testing of the application.

## 1. Introduction

Time management is one of the most challenging activities that students face daily, especially with tight deadlines and multiple subjects to prepare for. Most students are not able to prepare effectively for all their subjects because they usually allocate most of their time to subjects with higher weightage and skip those with lower weightage. This project focuses on developing a minimal viable planner application that aims to turn the subjective judgments of students about how difficult each subject is and its overall importance into objective facts about how much time should be devoted to studying the subject.

### 1.1 Problem Statement

Given a set of subjects with difficulty levels, days left for examination, and grade weightage, develop:

- a. a ranking of the subjects based on their relative priority,
- b. an algorithm to distribute a given amount of study hours among the subjects according to the calculated priority.

### 1.2 Objectives

After completing this project, the following should have been achieved:

1. A prioritization formula that quantifies the relative priority of each subject.
2. An interactive option to add or remove subjects as needed.
3. The subjects should be listed according to their calculated priority.
4. The study hours should be distributed among the subjects according to the relative priority.
5. Core java concepts such as classes and objects, collection framework, and control statements should be applied.

## 2. System Requirements

### 2.1 Software Requirements

The following software tools are required to implement the project:

1. Java Development Kit (JDK) version 8 or higher
2. Text editor or IDE
3. Operating System: Windows, macOS or Linux or any other OS that supports Java

### 2.2 Hardware Requirements

The following hardware components are required to run the application:

1. CPU: 2 GB RAM or higher
2. Computer: Any desktop or laptop computer that can support the Java Virtual Machine

### 3. System Design

#### 3.1 Class Design

The following classes were designed for the project:

| Class Name        | Description                                                                                                                                                                                          |
|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Subject           | This class stores the subject name, difficulty level, days left before examination, and the credits for the subject. It also contains the method for calculating the subject's priority score.       |
| StudyPlanner      | This class tracks the list of subjects. It also contains the methods for adding or removing a subject, calculating the relative priority score, and distributing the study hours among the subjects. |
| SmartStudyPlanner | This is the primary class for the application. It is responsible for displaying the menu options and receiving user input.                                                                           |

#### 3.2 Data Flow

The user interacts with the Main class through the console to add or remove a subject. The data about the subject such as the name, difficulty level, days left for the examination, and credits are used to create a subject instance. These instances are stored in an ArrayList object belonging to the StudyPlanner instance. When the user chooses to view the priority rank or study time allocation, the StudyPlanner goes through the list of subjects, calculates and returns their priority scores, and displays the result.

### 4. Algorithm / Priority Formula

The algorithm used to calculate the subject priority is as follows:

Urgency =  $100 / (\text{Days Left} + 1)$   
Priority Score =  $(\text{Difficulty} \times 2) + \text{Urgency} + (\text{Credits} \times 1.5)$

#### Explanation

- The difficulty level is multiplied by two to show that more time should be devoted to subjects that are more difficult. This value is adjusted depending on the number of days left before the examination.
- The urgency is highest when the days left for the examination are lowest, hence the inverse relationship. On the day of the examination, the urgency reaches its maximum value of 100. The +1 in the formula ensures that dividing by zero days does not occur on the day of the exam.
- The credits are multiplied by 1.5 to emphasize their importance while not overemphasizing them compared to the difficulty level.
- Once the priority score for each subject has been determined, the next step is to allocate the study hours for the day by dividing the total number of hours in proportion to each subject's priority score.

### 5. Implementation

#### 5.1 Data Structures Used

The following data structures were used to implement the project:

1. ArrayList - used to store the dynamically added subjects.
2. Comparator - used to define the custom sort order for comparing objects.

## 5.2 Key Methods Used

1. `addSubject()` - This method creates a new subject and calculates its priority score.
2. `removeSubject()` - This method deletes an existing subject instance.
3. `viewRanking()` - This method displays the subject ranking according to their computed priority score.
4. `allocateHours()` - This method calculates the total hours for the day and distributes them in proportion to the priority scores.

## 5.3 Sample Menu

```
===== SMART STUDY PLANNER =====
1. Add Subject
2. Remove Subject
3. View Priority Ranking
4. Generate Today's Study Plan (allocate hours)
5. Exit
```

## 6. Testing

The following test cases were carried out to verify the correctness of the project:

| Test Case                   | Description                                                                           | Expected Result                                                             | Outcome |
|-----------------------------|---------------------------------------------------------------------------------------|-----------------------------------------------------------------------------|---------|
| Add subject                 | Adding a subject with a name, difficulty level of 8, days left of 5, and credits of 4 | The subject should appear in the ranking with the calculated priority score | Pass    |
| Urgent exam                 | Adding a subject with days left of 0                                                  | Urgency score should be 100 (highest)                                       | Pass    |
| Remove subject              | Removing an existing subject                                                          | The subject should be deleted from the list                                 | Pass    |
| Remove non-existent subject | Removing a subject with a name that is not in the list                                | An error message "Subject not found" should be displayed                    | Pass    |
| Hour allocation             | Allocating 6 hours for multiple subjects                                              | The sum of the allocated hours for all subjects should equal 6 hours        | Pass    |
| Invalid menu option         | Entering a non-numeric menu option                                                    | The option should be rejected, and an error message displayed               | Pass    |

## 7. Output Screenshots

*[Insert screenshots here]*

## 8. Conclusion

The Smart Study Planner application is a good example of how a simple prioritization formula can be used to determine the relative importance of different activities and allocate limited resources based on that. Through this project, several aspects of core Java programming such as the collection framework, comparator, scanner, and console I/O were mastered. It was also satisfying to develop a useful application that would help other students manage their time more efficiently.

## 9. Future Scope

The scope of this project can be expanded by:

1. Storing the subjects in a file or database so that the data is persistent.
2. Adding a Graphical User Interface (GUI) to make the application more user-friendly.
3. Enabling the user to configure the weights used in the priority calculation formula.
4. Providing reminders to the user as the examination date approaches.

5. Adding options to track and compare actual study hours with planned hours.

## **10. References**

1. Oracle Java SE Documentation – <https://docs.oracle.com/en/java/javase/>
2. Documentation for ArrayList and Collections framework
3. Course lecture notes on Object Oriented Programming Concepts in Java